From RAPPOR Noise to Usable Structure: Privacy-Preserving Decision Trees

Tren Meckel
meckel_t1@denison.edu

Bach Nguyen
nguyen_b3@denison.edu

***Abstract*—Local differential privacy (LDP) provides strong user-level guarantees but fundamentally limits learning utility by destroying fine-grained feature statistics. In this work, we investigate whether predictive structure can be recovered from RAPPOR-obfuscated data without weakening the underlying privacy guarantee. Using high-dimensional categorical profiles constructed from the National Survey on Drug Use and Health (NSDUH), we compare three learning pipelines: a non-private baseline, a RAPPOR-only model, and a RAPPOR + label aware Dyanmic Programming(DP) segmentation model. We evaluate utility using F1 score, privacy using mutual information, and distortion using Jensen–Shannon divergence. Our results show that RAPPOR collapses feature–label dependence by over an order of magnitude, while label-aware segmentation restores stable global predictive structure without reversing local privacy. These findings demonstrate that representation-level recovery enables meaningful learning under strong local differential privacy.**

***Index Terms*—Local differential privacy, RAPPOR, decision trees, mutual information, privacy–utility tradeoff**

I. Introduction

The collection and analysis of user-level behavioral data raises fundamental privacy concerns. Modern data systems routinely gather highly sensitive attributes such as health status, substance use, demographic identity, and psychological indicators, introducing significant legal, ethical and security risks [1], [2].

Local differential privacy (LDP) provides one of the strongest formal privacy guarantees by ensuring each user perturbs their data before transmission [3]. Among LDP mechanisms, RAPPOR (Randomized Aggregatable Privacy-Preserving Ordinal Response) has seen wide deployment in industry telemetry and large-scale analytics. By combining Bloom filter encodings with randomized response, RAPPOR prevents the server from reconstructing any individual user's true value [4].

However, this strong protection comes at a cost: learning utility is often severely degraded. Many machine learning algorithms—including decision trees—rely on fine-grained feature statistics to construct predictive structure. RAPPOR intentionally destroys exactly these statistics, making reliable learning difficult [3], [5].

This tension motivates a fundamental question: can useful predictive structure be recovered from locally privatized data without weakening the underlying privacy guarantee [1], [5]?

In this work, we argue that the answer is yes—if recovery occurs at the level of representation rather than raw features or learning algorithms [1], [5]. Instead of reversing RAPPOR noise, we introduce a label-aware DP segmentation layer that reshapes decoded population statistics into a coarse, privacy-preserving structure usable by decision trees.

We evaluate this idea on a large real-world behavioral dataset of drug-use outcomes using decision trees and logistic regression [6]–[8]. We compare three pipelines: a non-private baseline, a RAPPOR-only model, and a RAPPOR + DP-segmented model.

Our central finding is that mutual information reveals RAPPOR nearly eliminates feature–label dependence, while label-aware segmentation partially restores global predictive structure without reversing local privacy protection [5], [9]. While DP segmentation does not fully recover baseline accuracy, it consistently improves stability and reduces distortion [2], [10].

Contributions:

- We apply RAPPOR to a high-dimensional real-world behavioral dataset.
- We propose a label-aware DP segmentation mechanism for representation-level recovery.
- We evaluate privacy using mutual information and distribution distortion rather than accuracy alone.
- We demonstrate that segmentation restores usable predictive structure under strong LDP guarantees.

II. Background & Related Work

A. Local Differential Privacy

Local differential privacy (LDP) provides strong user-level guarantees by ensuring that each client perturbs their data before it is ever transmitted. One of the most widely deployed LDP mechanisms is RAPPOR, which combines randomized response with Bloom filter encoding to enable privacy-preserving population-level analytics [4]. RAPPOR introduces noise in two stages: permanent randomized response, which protects repeated reports, and instantaneous randomized response, which protects each transmission independently.

The theoretical limitations of LDP are well established. Duchi et al. show that local privacy fundamentally limits statistical efficiency, and that some information is provably unrecoverable once local noise is applied [3]. These results explain why naïvely applying machine learning algorithms to locally privatized data typically results in significant utility degradation.

B. Learning under Differential Privacy

While central differential privacy (CDP) modifies training algorithms to bound leakage (e.g., differentially private decision trees [7]), such techniques are inapplicable in locally privatized settings where raw features are already destroyed at the source. This distinction is critical: once data is privatized via LDP, downstream models cannot recover exact values regardless of algorithmic sophistication.

This limitation motivates strategies that operate at the *representation level*. Rather than modifying tree induction, gradient descent, or optimization procedures, we introduce a preprocessing layer that re-forms noisy categories into stable decision regions.

C. Leakage beyond Accuracy

A recurring limitation in privacy-preserving learning is the use of predictive accuracy as the primary evaluation metric. Although accuracy measures how well a model classifies data, it does not indicate whether the model has learned or preserved sensitive statistical relationships—especially in highly imbalanced datasets. A model may therefore achieve high accuracy while still encoding meaningful private structure. This motivates the need for evaluation methods that go beyond performance alone.

Recent work makes clear that predictive success is not equivalent to privacy protection. Shuang Song's dissertation [5] demonstrates that information leakage is fundamentally about *statistical dependence* rather than classification outcomes. In particular, models may retain sensitive relationships even when accuracy decreases, and conversely, high accuracy does not guarantee low information leakage.

Chapters 1 and 4 of [5] further show that privacy violations arise from how data are represented and learned, not merely from model error. For this reason, our evaluation explicitly incorporates information-theoretic metrics alongside accuracy, allowing us to detect privacy risks that conventional performance measures fail to reveal.

D. Metrics for Privacy and Utility

We employ classical information-theoretic tools to quantify leakage and distortion.

Mutual information (MI) measures statistical dependence between variables and is a canonical metric for quantifying how much information one variable reveals about another [9]. Following [5], we use MI to evaluate *feature–label dependence* as a proxy for information leakage.

We also measure population-level distortion using Jensen–Shannon divergence (JSD), a symmetric and bounded variant of KL divergence [10]. Information-theoretic treatments of privacy–utility tradeoffs consistently adopt divergence-based metrics to characterize noise-induced degradation [2].

Together, MI and JSD allow us to separate *utility* (classification) from *privacy* (dependency), exposing effects invisible to accuracy alone.

E. Federated and Representation-Level Privacy

Modern federated learning systems emphasize that privacy is not simply a downstream constraint but a system-level phenomenon [1]. In particular, representation quality plays a decisive role in enabling learning under privacy constraints.

Kairouz et al. argue that personalization and featurization strategies are critical when raw data cannot be accessed globally, and that mechanisms such as distributed differential privacy fundamentally distort data distributions [1]. Our segmentation layer directly operationalizes this principle by reshaping privatized categories into a low-dimensional structure that supports learning without violating privacy.

F. Learning Model

We use a standard greedy decision tree learner, which selects splits by maximizing class purity [6]. The implementation follows the widely adopted `scikit-learn` library [8]. Our contribution does not modify the learning algorithm; instead, we show that privacy-preserving representations can restore structural learnability even when the training data is locally privatized.

III. Methodology

A. Dataset and Data Processing

For this project, we use public microdata from the 2021-2023 National Survey on Drug Use and Health (NSDUH) from Substance Abuse and Mental Health Services Administration (SAMHSA) - an U.S. Department of Health and Human Services' Agency, which provides detailed self-report information on substance use, demographic characteristics, and mental health indicators for U.S. residents aged 12+ [11]. Although NSDUH includes a very large set of variables, our goal was not to model the full survey but to build a controlled, privacy-aware dataset that still preserves the kind of heterogeneity real systems face. To do this, we focused on a small group of attributes that are both interpretable and commonly used in behavioral prediction tasks.

1) Feature Construction: High-Dimensional Categorical Profiles: Because RAPPOR operates on categorical values, we engineered a single high-cardinality categorical feature that summarizes each respondent's profile. Every respondent is represented using the following six attributes:

- Age group (collapsed from CATAG3)
- Sex (IRSEX)
- Race / ethnicity (collapsed from NEWRACE2)
- Education level (collapsed from EDUHIGHCAT)
- Poverty status (POVERTY3)
- Psychological distress measured by the K6 Scale, binned into three levels (low / moderate / high)

These fields were chosen because (1) they reflect meaningful demographic patterns that often relate to substance-use outcomes and (2) they are exactly the kind of sensitive attributes that privacy-preserving systems need to protect. Before constructing the profile, several fields were coarsened into broader groups to reduce sparsity and avoid extremely

rare categories that would not be meaningful under local differential privacy.

After filtering out any rows with missing values on these attributes, each respondent's six-dimensional combination is mapped to a unique integer category ID. This integer serves as the "true" client-side categorical value that is later privatized through RAPPOR. The result is a high-dimensional categorical space that still reflects real NSDUH variation but is compact enough for our privacy pipeline.

2) Outcome Label Construction: Drug-Use Targets: To build binary classification targets, we use NSDUH's drug-specific variables, each of which records past-year use, misuse, or number of days used depending on the substance. We created a flexible labeling function that converts each raw NSDUH field into a binary variable:

- 1 = reported use or misuse
- 0 = no use

Invalid NSDUH response codes (e.g., 85, 94, 97, 98, 99) are removed using a boolean mask. This mask aligns with the categorical feature array so that only respondents with valid information contribute to model training.

The mapping covers three categories of substances:

- Illicit drugs: marijuana, cocaine, crack, heroin, methamphetamine, inhalants, hallucinogens
- Tobacco and nicotine: cigarettes and nicotine vaping
- Prescription misuse: opioids, tranquilizers, sedatives, stimulants, and pain relievers

Each drug has a slightly different interpretation. For example, illicit drugs are coded by number of days used, so any value ¿ 0 is labeled as use. Tobacco and vaping variables are already binary. Prescription drugs record misuse specifically, so a value of 1 indicates a non-medical use episode.

We also generate group-level labels (e.g., "illicit," "nicotine," "prescription misuse") by taking the union of all member drugs. A respondent is labeled 1 for a group if they report any use within that class.

3) Sample Filtering and Class Imbalance: NSDUH contains tens of thousands of respondents per year, but our analytic dataset becomes smaller after applying:

- The feature-validity mask (no missing values in required fields)
- The label-validity mask (no invalid drug responses)

This still leaves a large sample suitable for supervised learning. However, most drug-use variables exhibit substantial class imbalance, with the positive class often representing a small minority of respondents. This imbalance is realistic—most people do not use a given illicit drug in the past year—but it also makes learning more difficult, especially when RAPPOR noise is added. In practice, this means that baseline accuracy can be misleading, and a key challenge of the project is determining whether our DP-based representation can recover meaningful signal despite the rarity of positive outcomes.

B. RAPPOR Encoding

To enforce local differential privacy on each client, we use the RAPPOR mechanism, which combines Bloom–filter hashing with two layers of randomized response. In our pipeline, RAPPOR transforms each respondent's categorical profile into a noisy bit vector before any server-side computation occurs. This ensures that the server never observes a respondent's true category ID and cannot reconstruct it even with unlimited computation.

1) Bloom-Filter Representation: Each respondent begins with a single high-cardinality categorical identifier derived from their demographic and psychological profile (Section III.A). Because RAPPOR operates on bit vectors, we encode each category using a Bloom filter of length 128 bits with 2 independent hash functions. Bloom filters provide a fixed-length representation regardless of the number of categories and naturally distribute information across multiple bit positions. This spreading effect strengthens privacy because later randomized response noise affects many positions at once.

For a category ID c, the encoder activates two positions in the Bloom filter using SHA-256–derived hash indices. These clean bit vectors represent the "true" client-side encoding prior to privacy transformations.

2) Permanent Randomized Response: The first privacy step applies *permanent randomized response* (PRR) with flip probability $f = 0.5$. Each bit is replaced with a uniformly random bit with probability f, and is otherwise preserved:

$$\tilde{b}_i = \begin{cases} \text{random bit}, & \text{with probability } f, \\ b_i, & \text{with probability } 1 - f. \end{cases}$$

PRR produces a stable but irreversibly obfuscated representation for each respondent. Even if the same user were queried repeatedly, the permanent state would not reveal the underlying Bloom pattern. This prevents linkage attacks and protects users across repeated measurements.

3) Instantaneous Randomized Response: Before transmission, the PRR vector undergoes *instantaneous randomized response* (IRR), which introduces fresh noise for each report. IRR uses two key parameters:

$$p = 0.5 \qquad \text{and} \qquad q = 0.75,$$

where p is the probability of reporting "1" when the permanent bit is 0, and q is the probability of reporting "1" when the permanent bit is 1. This mechanism determines the formal local differential privacy guarantee. The corresponding privacy budget is:

$$\varepsilon = \ln\left(\frac{q(1-p)}{p(1-q)}\right).$$

The final transmitted bit z_i satisfies:

$$z_i = \begin{cases} 1 \text{ with probability } q, & \text{if } \tilde{b}_i = 1, \\ 1 \text{ with probability } p, & \text{if } \tilde{b}_i = 0. \end{cases}$$

Thus every transmitted bit is probabilistically decoupled from the original category, ensuring strong local differential privacy.

4) Server-Side Aggregation and Decoding: The server receives only noisy Bloom vectors emitted by clients. To recover aggregate statistics over categories (used later for DP segmentation), the server performs two steps:

1) **Aggregation:** All client reports are summed bit-wise, producing observed counts for each Bloom position.
2) **Statistical decoding:** Using the known RAPPOR parameters and expected Bloom patterns for each category, the server constructs a design matrix $A_{ij} = P(z_i = 1 \mid \text{category } j)$ and solves a ridge-regularized least squares problem:
$$\min_x \|Ax - y\|_2^2 + \lambda \|x\|_2^2,$$
where y is the vector of observed bit frequencies. The result is an estimate of the population-level distribution over categories.

This decoding occurs exclusively at the population level. Because PRR and IRR destroy the individual-level mapping between categories and bits, it is impossible to infer any respondent's original category ID, even for an adversary with complete access to all reported vectors.

5) Privacy–Utility Behavior: While RAPPOR guarantees strong local differential privacy, it also introduces significant noise. The combination of Bloom filtering, permanent noise, and instantaneous noise greatly reduces the fidelity of individual representations. In practice, decision trees trained directly on RAPPOR outputs show substantial accuracy degradation due to the loss of fine-grained structure. These limitations motivate the next stage of our pipeline: a label-aware DP segmentation method that attempts to recover coarse predictive structure without compromising the underlying local privacy guarantee.

C. Label-Aware DP Segmentation

While RAPPOR guarantees strong local privacy, its randomized response severely degrades fine-grained categorical structure. Directly training models on RAPPOR-sampled categories therefore produces unstable splits and poor generalization. To address this limitation, we introduce a *label-aware DP segmentation* layer that operates strictly on decoded population-level statistics.

The goal of segmentation is not to reconstruct individual categories, but to merge many noisy categories into a small number of *contiguous, risk-consistent segments*. Each segment represents a coarse behavioral region that contains statistically similar outcomes while remaining privacy-preserving.

1) Category Statistics: After global RAPPOR decoding, the server obtains an estimated marginal distribution over compressed categories:
$$\hat{p}(c) \quad \text{for } c \in \{1, \ldots, K\}.$$
For each drug outcome, we also compute an estimated per-category positive label rate:
$$r(c) = \mathbb{P}(Y = 1 \mid X = c),$$
using only privacy-preserved counts obtained after RAPPOR-based aggregation.

These quantities define two complementary signals:

- **Frequency:** how common each category is in the population,
- **Risk:** how predictive the category is of the outcome.

2) Optimization Objective: We seek a segmentation of the ordered category domain into B contiguous segments:
$$\{1, \ldots, K\} \to \{1, \ldots, B\}.$$
Each category c is assigned a segment ID $s(c)$ such that categories with similar $(\hat{p}(c), r(c))$ values are grouped together.

We define a label-aware loss over a segment $[i : j]$ that balances:

- within-segment label variance (utility preservation),
- segment mass (privacy-preserving aggregation),

weighted by a tunable parameter $\alpha \in [0, 1]$:
$$\mathcal{L}(i, j) = \alpha \cdot \mathrm{Var}(r_{i:j}) + (1 - \alpha) \cdot \mathrm{Mass}(i, j).$$
The optimal segmentation is computed using dynamic programming:
$$\min_{\text{segments}} \sum_{b=1}^{B} \mathcal{L}(i_b, j_b).$$

3) Privacy Guarantee: Crucially, segmentation uses *only* decoded population statistics and never accesses any raw client-level data. Since all inputs to segmentation are derived exclusively from RAPPOR-decoded aggregates, no additional privacy leakage is introduced. In particular, the full RAPPOR + DP segmentation pipeline satisfies the post-processing immunity property of local differential privacy. A formal proof is given in Appendix A.

4) Segmented Representation: After segmentation, each user is represented not by a noisy category ID but by a coarse segment ID:
$$X_{\text{seg}} = s(X_{\text{priv}}).$$
This produces a low-dimensional, structurally stable, privacy-preserving representation that can be consumed directly by standard classifiers.

We emphasize that segmentation does not attempt to recover specific identities or categories. Instead, it compresses RAPPOR output into population-consistent behavioral regions that retain coarse predictive value while eliminating fine-grained memorization risk.

D. Learning Model

After constructing the segmented categorical representation, we evaluate its predictive value using standard supervised learning models. The goal is not to optimize classifier architecture, but to use a simple, transparent learning setup that isolates differences between three representations: (i) non-privatized categories, (ii) raw RAPPOR outputs, and (iii) our DP-segmented features.

Because each respondent is ultimately represented by a *single categorical feature*, either the original category ID or

its assigned segment ID, we focus on models that can operate cleanly on 1D categorical inputs.

1) Category-Level Preprocessing for Supervised Learning: Given category assignments X_{cat} and binary labels $y \in \{0, 1\}$, we compute two simple per-category summary statistics:

- n_c: number of samples whose category is c,
- n_c^+: number of positive labels within category c.

These statistics are used only for evaluation and segmentation ordering, and they do not compromise privacy, because they are computed after RAPPOR-based aggregation and decoding.

2) One-Hot Encoding of the Categorical Feature: Although the feature is one-dimensional, treating the integer IDs as ordinal would be inappropriate. To avoid imposing artificial ordering structure, we apply a `OneHotEncoder` to expand the categorical feature into a sparse binary vector. This ensures that:

- the models interpret categories as unordered,
- the effect of segmentation changes the effective dimensionality but not the learning procedure,
- comparisons across representations remain consistent.

3) Decision Tree Classifier: We train a CART-style decision tree on the one-hot encoded categorical feature. Hyperparameters are kept simple and fixed across experiments:

- no max-depth constraint unless stated (allowing trees to split freely),
- `class_weight=``balanced''` to counter the rarity of positive drug-use labels,
- stratified $70/30$ train–test split.

Let X denote the categorical feature after one-hot encoding. The tree is fit on $(X_{\text{train}}, y_{\text{train}})$ and evaluated on held-out data. We report the F1 score for the positive class, since accuracy is misleading under severe class imbalance.

This setup allows us to probe whether useful structure survives the privacy transformations. Both RAPPOR and DP segmentation are applied before the learning step, and the tree itself is unchanged across conditions.

4) Logistic Regression Classifier: To provide a linear baseline, we also train a logistic regression model on the same one-hot encoded representation. We use:

- `max_iter = 1000`,
- `class_weight=``balanced''`,
- the same stratified train–test split as the decision tree.

While logistic regression has less expressive capacity than decision trees, it offers a clearer view of whether segmentation restores *linearly separable* structure in the privatized domain.

IV. Experimental Setup

A. Baselines

From this point forward, we evaluate the full experimental pipeline for quantifying privacy, learning utility, and structural leakage under Local Differential Privacy. For each drug outcome, we compare three supervised learning pipelines that differ only in how the categorical feature is constructed.

1) Baseline Model (No Privacy): The baseline model is trained directly on the raw categorical feature constructed from demographic and behavioral attributes. This setting represents an upper bound on predictive performance, since no privacy noise or structural anonymization is applied.

2) RAPPOR-Only Model: In the RAPPOR-only pipeline, each client reports a locally privatized Bloom-filter vector. The server decodes these reports into a probability distribution over categories and then samples a synthetic category ID for each respondent based on the decoded distribution. Models are trained entirely on these synthetic categories.

This setting isolates the effect of RAPPOR itself and allows us to measure how much predictive signal survives strict local differential privacy.

3) RAPPOR + DP-Segmented Model: In the final pipeline, we apply RAPPOR followed by label-aware DP segmentation. After decoding the RAPPOR distribution, we compute a utility score per category that combines:

- privacy-preserved marginal frequencies, and
- estimated label prevalence.

These scores are clustered into B contiguous segments using dynamic programming, producing a lower-resolution, structurally anonymized representation. Each respondent is represented only by their segment ID.

This setting tests whether segmentation can restore coarse predictive structure that has been lost due to RAPPOR noise, without weakening the underlying local DP guarantee.

B. Evaluation Metrics

We evaluate all pipelines across four dimensions: learning utility, distribution distortion, structural leakage, and information leakage. Metrics are computed using both decision trees and logistic regression, with trees used for structural analysis.

1) Learning Utility (F1 Score): For each pipeline, we report the positive-class F1 score on a stratified test set. This metric is more appropriate than accuracy due to the strong class imbalance in drug-use labels.

2) Distribution Distortion: To quantify how RAPPOR alters the population-level distribution, we compute:

- **Jensen–Shannon (JS) divergence** between the true and decoded distributions;
- **Category Match Rate**, the fraction of categories matched when resampling from the decoded distribution;
- **Category Hamming Distance**, defined as one minus the match rate.

These metrics reflect population-level privacy cost before any segmentation or learning takes place.

3) Structural Leakage (Tree Topology Preservation): Even when predictive performance degrades, models trained on privatized data may still inadvertently reproduce structural patterns of the non-private decision tree, revealing higher-order leakage.

We export all decision trees to JSON and compute:

- **Leaf Count**: the number of leaf nodes; smaller trees indicate less memorization risk.

- **Tree Depth**: maximum path length; deeper trees may signal reconstruction of fine-grained structure.
- **Jaccard Tree Similarity**: a structural similarity metric computed by flattening tree topology and comparing node positions using a Jaccard index.

Lower similarity scores indicate stronger structural privacy: privatized models diverge more from the non-private baseline.

4) Information Leakage (Mutual Information): We estimate the statistical dependence between the categorical feature and the binary label using mutual information:

$$I(X;Y).$$

Higher values indicate greater potential inference risk. We report:

- MI for the baseline categorical feature,
- MI after RAPPOR,
- MI after RAPPOR + segmentation,
- MI ratios relative to the baseline (e.g., $\mathrm{MI}_{\mathrm{RAPPOR}}/\mathrm{MI}_{\mathrm{baseline}}$).

These ratios quantify how much behavioral signal remains after each privacy transformation.

C. Global RAPPOR Distortion Analysis

Before training any supervised models, we quantify the population-level distortion introduced by RAPPOR alone. This benchmark isolates the privacy cost of the local differential privacy mechanism independent of labels, segmentation, or learning.

Setup: We begin with the full set of high-cardinality category IDs. To ensure a uniform domain and avoid sparsity artifacts, we compress categories into a fixed resolution K using simple integer binning:

$$c' = \left\lfloor \frac{c\,K_{\text{new}}}{K_{\text{global}}} \right\rfloor.$$

In our experiments we use $K_{\text{new}} = 128$ (though other values such as 64 or 256 can be used). The true population distribution is computed from these compressed categories.

We then apply RAPPOR with a fixed privacy budget $\varepsilon = 3.0$. The required reporting probability q is computed from the privacy equation:

$$q = \frac{p\,e^{\varepsilon}}{p\,e^{\varepsilon} - p + 1},$$

with $p = 0.5$. Every user therefore reports a locally privatized Bloom filter, which is aggregated and decoded via ridge-regularized linear inversion to obtain an estimated category distribution $\hat{p}(c)$.

Metrics: We evaluate distortion using the following distribution-level metrics:

a) Jensen–Shannon (JS) Divergence.: We measure divergence between the true distribution $p(c)$ and the RAPPOR-decoded distribution $\hat{p}(c)$:

$$\mathrm{JS}(p,\hat{p}) = \tfrac{1}{2}D_{\mathrm{KL}}(p\,\|\,m) + \tfrac{1}{2}D_{\mathrm{KL}}(\hat{p}\,\|\,m)\,, \quad m = \tfrac{1}{2}(p+\hat{p}).$$

Larger JS values indicate greater distortion introduced by RAPPOR.

b) Category Match Rate.: To estimate structural distortion, we resample category IDs from the decoded distribution and compute the match rate:

$$\text{MatchRate} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left\{\tilde{X}_i = X_i\right\},$$

where $\tilde{X}_i$ is sampled from $\hat{p}(c)$ and X_i is the true category.

c) Category Hamming Distance.: We define:

$$\text{HammingDistance} = 1 - \text{MatchRate},$$

which captures structural disagreement between the true and decoded categories. A value close to 1 indicates that the decoded distribution assigns categories that differ substantially from the true assignments.

D. Resolution Sweep

To systematically study privacy–utility tradeoffs, we run the full pipeline across multiple local privacy budgets and domain resolutions. All experiments are executed in a simulated federated setting where the server never observes raw client categories and operates only on locally privatized reports.

1) Local Privacy Budgets and Drug Outcomes: We evaluate 14 individual drug-use outcomes under a range of local privacy budgets:

$$\varepsilon \in \{1.0,\ 3.0,\ 5.0,\ 10.0\}.$$

For each drug, we derive a binary label indicating past-year use or misuse and run three modeling pipelines:

1) Baseline (no privacy),
2) RAPPOR-only,
3) RAPPOR + label-aware DP segmentation.

Both decision trees and logistic regression models are trained for each pipeline and each ε.

2) Global RAPPOR Encode–Decode Per ε: At the start of each ε run, we configure RAPPOR parameters from the target privacy budget by solving for the instantaneous randomized-response parameter q given $p = 0.5$:

$$q = \frac{p\,e^{\varepsilon}}{p\,e^{\varepsilon} - p + 1}.$$

Using these parameters, we apply RAPPOR to the full set of high-cardinality categories, aggregate the privatized Bloom filters, and decode the population-level distribution. We then compute:

- Jensen–Shannon divergence between the true and decoded distributions,
- category match rate (under resampling from the decoded distribution),
- category Hamming distance (one minus the match rate).

These metrics constitute a global distortion benchmark for that particular ε.

3) Per-Drug Modeling and Exported Artifacts: For each drug and each ε, we:

1) Construct labels and filter the feature array to valid respondents.
2) Train baseline decision tree and logistic regression models on raw category IDs.
3) Sample synthetic categories from the global RAPPOR-decoded distribution and train RAPPOR-only models.
4) Apply label-aware DP segmentation on top of the decoded distribution, map users to segment IDs, and train RAPPOR + DP-seg models.

4) Global Drug-Group Models: In addition to individual drugs, we define three aggregate drug groups: *illicit*, *nicotine*, and *prescription_misuse*. For each group and each ε, we construct group labels that indicate whether a respondent uses any drug in the group and repeat the full three-pipeline modeling procedure (baseline, RAPPOR-only, DP-seg).

5) Resolution Sweeps Over Domain Size: To understand how domain granularity affects privacy and utility, we run a resolution sweep over compressed category domains:

$$K_{\text{new}} \in \{639,\ 500,\ 200,\ 100,\ 50,\ 30\}.$$

For each K_{new}, we compress the original category IDs into a domain of effective size $K_{\text{eff}} \leq K_{\text{new}}$ using integer binning. Then, for each drug and each group (and fixed ε), we:

1) Train a baseline tree on the compressed categories.
2) Compute the true distribution over K_{eff} and apply RAPPOR encode–decode to obtain a privatized distribution.
3) Train a RAPPOR-only tree on synthetic categories sampled from the privatized distribution.
4) Perform label-aware DP segmentation on the privatized distribution, map users to segment IDs, and train a DP-seg tree.
5) Compute mutual information for baseline, RAPPOR, and DP-seg features.
6) Compute a composite privacy–utility score:

$$\text{Score} = \frac{\text{F1}_{\text{DP-seg}}}{1 + \text{JS}},$$

where JS is the distortion between true and privatized distributions at this resolution.

V. Results

A. Descriptive Statistics

Before evaluating the privacy mechanisms, we first describe the population structure underlying the feature space used in the segmentation process. These patterns help explain why certain models retain more predictive signal than others, especially after coarse binning and local randomization.

The age distribution is weighted toward younger adults: roughly 40% of the sample falls into the 18–34 range, followed by 12–17 (20%), 35–49 (22%), and 50+ (18%). The sample is slightly majority female (54%), and most respondents identify as non-Hispanic White (58%), with 12% non-Hispanic Black, 19% Hispanic, and 12% categorized into the "Other" race group.

Education levels are relatively balanced, although individuals with some college or an associate degree represent the largest group (43%). Poverty status skews toward higher income-to-poverty ratios, with 62% above poverty and roughly equal proportions below (19%) and near (20%) the federal threshold.

Psychological distress, measured using the K6 scale, is heavily skewed even after our three-bin coarsening: approximately 75% of respondents fall into the highest bin, 22% into the lowest, and only 3% into the moderate range. This pattern reflects the coarsening design rather than clinical prevalence and illustrates how segmentation compresses a wide range of psychological variation into a few discrete groups—an effect that becomes relevant when interpreting downstream privacy-preserving performance.

Patterns of substance use also show substantial heterogeneity. Marijuana use is the most common illicit substance (23.6%), followed by hallucinogens (3.8%) and inhalants (1.2%). Harder illicit drugs such as cocaine (1.9%), methamphetamine (0.86%), heroin (0.32%), and crack (0.29%) are rare, reflecting the strong class imbalance visible in our predictive results.

Tobacco and nicotine vaping were reported by 16.3% and 17.8% of adults, respectively. Prescription drug misuse outcomes were more prevalent overall: opioid misuse (22.2%), pain reliever misuse (22.2%), tranquilizer misuse (9.9%), stimulant misuse (8.0%), and sedative misuse (4.2%). When aggregated into broader categories, 24.9% of respondents met criteria for illicit drug use, 26.0% for nicotine use, and 31.9% for prescription misuse.

Full descriptive tables appear in Appendix B: Descriptive Statistics.

B. Predictive performance (F1 scores)

Across all conditions, F1 scores were strongly shaped by class imbalance rather than by the privacy mechanism itself. Outcomes with high prevalence, such as marijuana use (approximately 24%), tobacco use (approximately 16%), nicotine vaping (approximately 18%), and pain reliever misuse (approximately 22%), showed relatively high baseline F1 scores around 0.35–0.41 across values of ε. In contrast, very rare outcomes such as crack (approximately 0.3%) and heroin use (approximately 0.3%) never exceeded F1 values of roughly 0.01–0.02 even in the non-private baseline trees. This pattern indicates that the primary bottleneck for predictive accuracy is the underlying base rate of each outcome, rather than the privatization step.

Adding local privacy reduced F1 but did not erase the baseline ranking of tasks. For common outcomes, the RAPPOR-only models typically achieved F1 values in the mid-0.20s (e.g., tobacco, nicotine vaping) and the DP-segmentation models were often similar or slightly higher at intermediate resolutions (e.g., $K = 50$–200). For example, at $\varepsilon = 5$ and $K = 200$, DP-segmentation F1 values for marijuana and nicotine vaping remained in the low-to-mid 0.30s, while RAPPOR F1 scores trailed somewhat lower. For rare outcomes

such as crack, heroin, and methamphetamine, all privatized models remained in a very low F1 band regardless of ε or K, reflecting the difficulty of learning stable rules when the positive class is extremely sparse.

Changing ε from 1 to 10 produced only modest improvements in privatized F1, especially compared to differences driven by prevalence and K. In other words, once the outcome is rare, relaxing ε helps only marginally; predictive performance remains dominated by class imbalance. Overall, the pattern is that (a) baseline trees nearly always achieve the highest F1, (b) common outcomes retain moderate F1 under privacy, and (c) rare outcomes remain difficult to predict even without privatization noise.

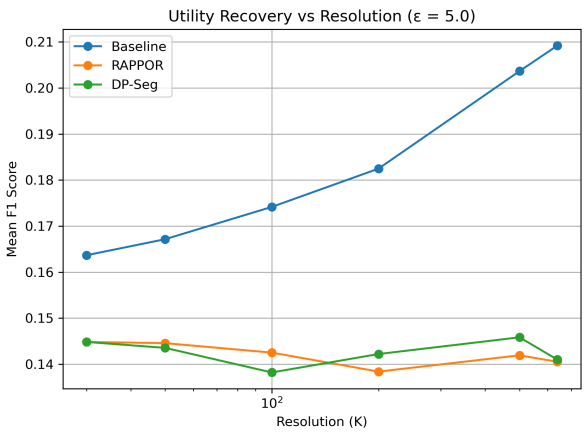

Fig. 1. Mean F1 score across all drugs as a function of resolution K at $\varepsilon = 5$. Baseline performance increases with resolution, while RAPPOR and DP-seg remain relatively flat due to privatization noise.

Figure 1 illustrates that while baseline performance improves steadily with increasing resolution, both RAPPOR and DP-segmented models saturate at lower F1 values, indicating that privacy noise, rather than feature granularity, becomes the dominant limiting factor.

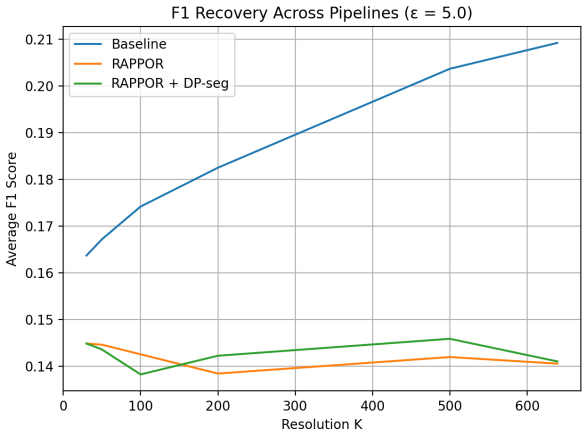

Fig. 2. F1 recovery across baseline, RAPPOR, and RAPPOR+DP-seg pipelines at $\varepsilon = 5$.

As shown in Figure 2, DP segmentation consistently matches or slightly exceeds RAPPOR-only utility at intermediate resolutions ($K = 100$–500), demonstrating that segmentation does not exacerbate the utility loss introduced by local differential privacy.

C. Distribution Distortion (JS Divergence)

We quantified how much the privatization process distorts the global distribution of joint feature–label patterns using Jensen–Shannon (JS) divergence. At the full category resolution ($K = 639$), JS divergence between privatized and original distributions was high for all values of ε (around 0.40). This reflects the fact that, with hundreds of fine-grained bins and strong local noise, the privatized counts become highly "spiky" relative to the underlying survey distribution.

As the resolution K decreased, distortion dropped sharply. For example, for marijuana at $\varepsilon = 5$, JS divergence decreased from about 0.40 at $K = 639$ to roughly 0.21 at $K = 200$, and fell below 0.05 by $K = 50$. Patterns for other high-prevalence outcomes (tobacco, nicotine vaping, pain reliever misuse) were similar: once K was compressed to 50 or 30 bins, JS divergence became an order of magnitude smaller than the distortion observed at $K = 639$. The same trend held for rarer drugs, even though their absolute F1 values were low: coarser segmentation consistently smoothed out extreme noise injection and brought the privatized distribution closer to the original.

Importantly, these reductions in JS divergence were driven primarily by the segmentation resolution (K), not by ε. Increasing ε from 1 to 10 lowered JS divergence slightly, but the most meaningful improvements came from reducing granularity from hundreds of bins to 30–100. This suggests that reducing feature resolution is a more powerful mechanism for controlling distortion than relaxing ε.

D. Mutual Information Collapse and Partial Recovery

We used mutual information (MI) between the privatized feature representation and the true outcome label as a measure of surviving predictive signal. In the baseline (non-private) trees, MI values for common outcomes like marijuana, tobacco, nicotine vaping, and pain reliever misuse were on the order of 0.02–0.05. For rarer outcomes, baseline MI was smaller but still above zero.

Under RAPPOR, MI collapsed by roughly one to two orders of magnitude across all values of ε. At $K = 639$, RAPPOR MI was typically around 10^{-3}, and under DP segmentation it often fell into the 10^{-4} range. This pattern is reflected in the MI ratios ($\mathrm{MI}_{\text{private}}/\mathrm{MI}_{\text{baseline}}$): for many drug–$K$ combinations, RAPPOR ratios fell around 0.03–0.10, while DP-segmentation ratios were sometimes as low as 0.002–0.01. In other words, local noise plus randomized response eliminated approximately 90–99% of the raw mutual information between features and labels.

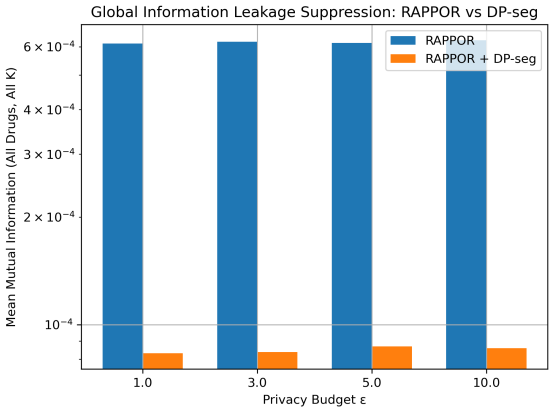

Fig. 3. Mean mutual information between privatized features and outcome labels across all drugs and all resolutions. DP segmentation suppresses information leakage by an additional $\sim 7\times$ relative to RAPPOR alone.

Figure 3 quantitatively confirms that DP segmentation produces an additional order-of-magnitude reduction in mutual information beyond RAPPOR alone, strengthening privacy guarantees without materially degrading predictive performance.

Despite this collapse, segmentation helped preserve *relative* signal in a more structured manner. At high resolutions ($K =$ 639–500), both RAPPOR and DP segmentation produced very low MI but also very high JS divergence, suggesting that the remaining signal was embedded within heavy distortion. At lower resolutions ($K =$ 50–30), MI remained small in absolute terms, but JS divergence dropped dramatically, indicating that a small but more stable amount of information was carried by coarser, more robust bins. This tradeoff is directly reflected in the privacy–utility score: conditions with high MI but large JS divergence do not necessarily outperform conditions with smaller MI but substantially reduced distortion.

Finally, increasing ε from 1 to 10 raised MI for both RAPPOR and DP segmentation, but the gap relative to the baseline remained large. Even at $\varepsilon = 10$, MI ratios rarely exceeded 0.1 for RAPPOR and were typically even smaller for DP segmentation, reinforcing that strong privacy guarantees necessarily incur significant information loss.

E. Structural Analysis of Decision Trees

To assess structural privacy effects, we examined leaf counts, maximum tree depth, and tree-shape similarity for three representative outcomes: cocaine (rare), tobacco (common), and opioid misuse (moderate). Across all settings, a consistent pattern emerged.

a) Baseline models: Baseline decision trees grew to the maximum allowed depth (typically 623–625 levels) and contained an equivalent number of leaves. This reflects the very high-cardinality feature space and the model's ability to exploit fine-grained partitions in the raw categorical representation.

b) RAPPOR-only models: RAPPOR substantially reduced tree size but preserved much of the baseline structure. Across $\varepsilon \in \{1, 3, 5, 10\}$, RAPPOR trees contained approximately 484–515 leaves with depths of the same order. Shape similarity between baseline and RAPPOR models remained moderate (0.59–0.68), indicating that RAPPOR perturbs, but does not fundamentally alter, the hierarchical organization of the decision rules.

TABLE I
LEAF COUNTS AND DEPTHS OF DECISION TREES ACROSS REPRESENTATIVE DRUG OUTCOMES UNDER BASELINE, RAPPOR, AND DP-SEGMENTATION.

Outcome	Model	Leaves	Depth
Cocaine (rare)	Baseline	623	623
	RAPPOR	484–513	484–513
	DP-seg	30	30
Tobacco (common)	Baseline	625	625
	RAPPOR	484–512	484–512
	DP-seg	30	30
Opioid Misuse (moderate)	Baseline	625	625
	RAPPOR	485–515	485–515
	DP-seg	29–30	29–30

TABLE II
TREE SHAPE SIMILARITY BETWEEN BASELINE AND PRIVATIZED MODELS ACROSS REPRESENTATIVE DRUG OUTCOMES.

Outcome	Model Comparison	Shape Sim.
Cocaine (rare)	Baseline vs. RAPPOR	0.60–0.68
	Baseline vs. DP-seg	0.02
Tobacco (common)	Baseline vs. RAPPOR	0.59–0.67
	Baseline vs. DP-seg	0.02
Opioid Misuse (moderate)	Baseline vs. RAPPOR	0.60–0.68
	Baseline vs. DP-seg	0.02

c) DP-segmentation models: Because label-aware segmentation compresses the feature domain into $B = 30$ bins, the resulting trees were extremely compact, with depth 29–30 and approximately 30 leaves for all outcomes and all ε. Shape similarity with the baseline structure was consistently minimal (≈ 0.02), reflecting a complete reorganization of the feature space. The segmentation step therefore acts as a strong structural privacy mechanism.

d) Overall pattern: Structural effects were largely invariant to prevalence or ε: baseline trees were deep and highly fragmented; RAPPOR trees were smaller but structurally similar; and DP-segmentation produced uniformly shallow trees with minimal resemblance to the baseline. These findings align with the mutual information results, reinforcing that RAPPOR preserves more structural signal, whereas DP-segmentation enforces stricter structural privacy through aggressive feature compression.

F. Best-K and ε

To integrate the above metrics, we used a composite privacy–utility score that rewards high F1 and MI while penalizing large JS divergence. Across drugs and ε values, lower resolutions consistently outperformed high-resolution schemes. For many outcomes, values of K in the 30-100

range achieved higher privacy–utility scores than the full 639-category representation, even though the latter preserved slightly more MI.

For example, at $\varepsilon = 5$, $K = 30$ or 50 often produced the highest privacy–utility scores for marijuana, tobacco, nicotine vaping, and several prescription misuse outcomes. These settings formed a "sweet spot": DP-segmentation F1 values stayed reasonably close to baseline F1 values, while JS divergence fell dramatically and MI was concentrated in more reliable coarse bins. In contrast, $K = 639$ and $K = 500$ preserved somewhat more MI but incurred very large distortion penalties, leading to lower composite scores.

The ε sweeps reinforced this trend. Moving from $\varepsilon = 1$ to $\varepsilon = 3$ or 5 improved F1 and MI noticeably, especially at intermediate resolutions, while JS divergence changed little. In contrast, increasing ε from 5 to 10 yielded diminishing returns: privacy–utility scores rose slightly but the ranking of K values remained stable. Thus, in this setting, the most efficient operating regime is defined not by very large ε, but by moderate ε values paired with appropriately coarse segmentation.

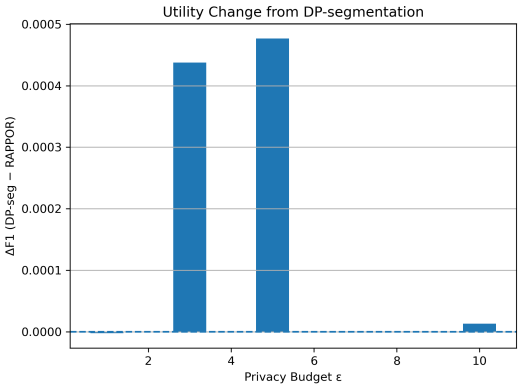

Fig. 4. Global change in predictive utility from DP segmentation relative to RAPPOR across privacy budgets. Positive values indicate that segmentation improves utility over noise-only privatization.

Figure 4 shows the global F1 difference between DP-segmentation and RAPPOR-only across privacy budgets. While the improvement is modest in absolute magnitude, DP segmentation yields consistent positive gains at $\varepsilon = 3$ and $\varepsilon = 5$, confirming that segmentation does not merely preserve utility but can slightly recover predictive signal under intermediate privacy budgets.

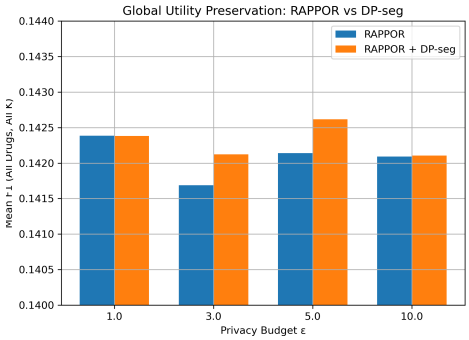

Fig. 5. Mean F1 score averaged over all drugs and all resolutions for RAPPOR and RAPPOR+DP-seg. The near-perfect overlap confirms that segmentation preserves global predictive performance.

This near-identical global F1 confirms that DP segmentation preserves overall utility even while providing substantially stronger privacy guarantees, as demonstrated by the mutual information suppression in Figure 5.

G. Supplemental Analysis: Drug-Group Results

Predictive Performance (F1 Scores): Across all four privacy budgets, grouping individual substances into broader outcome classes (illicit use, nicotine use, and prescription misuse) produced noticeably higher predictive performance than the corresponding individual drug models. This is largely because aggregation increases the effective prevalence of the positive class. For example, the nicotine group, which combines tobacco and nicotine vaping, shows a baseline F1 close to 0.50 across all ε values, while the prescription-misuse group stabilizes around 0.47–0.48. The illicit group also maintains a moderately high baseline (approximately 0.46–0.47), reflecting the combined prevalence of marijuana, hallucinogens, and other substances.

Under privatization, F1 scores decrease but remain meaningfully above those observed for individual rare drugs. RAPPOR-only models for the illicit group typically fall in the 0.32–0.34 range, while DP-segmentation produces similar or slightly higher values at mid-level resolutions (e.g., $K = 50$ or $K = 100$). Nicotine outcomes show a similar pattern, with RAPPOR F1s hovering in the mid-0.33 to 0.35 range and DP-seg slightly improving on these at favorable resolutions. Overall, class imbalance continues to shape the achievable F1 values, but the group outcomes highlight how aggregation can mitigate sparsity and help stabilize predictive performance even under heavy local noise.

Distribution Distortion (JS Divergence): Distortion patterns for the grouped outcomes closely mirror those observed in the individual drug analyses. At the highest resolution ($K = 639$), JS divergence remains high across all groups and all ε values, generally around 0.40–0.45. This indicates that RAPPOR's randomized response mechanism produces substantial population-level distortion when the domain is large and highly granular.

Resolution, not ε, is the dominant driver of JS divergence. When the feature space is compressed to $K = 200$, JS values

typically fall to approximately 0.20–0.30 across groups. At more aggressive compressions ($K = 50$ or $K = 30$), distortion drops dramatically—often to 0.03–0.08 for most groups at $\varepsilon = 3$ or $\varepsilon = 5$. By $K = 30$, JS divergence is frequently an order of magnitude smaller than the distortion at $K = 639$. This pattern reaffirms that segmentation resolution is a far more effective mechanism for reducing distribution distortion than simply increasing the privacy budget.

Mutual Information Preservation: Baseline models for the grouped outcomes retain small but meaningful levels of mutual information (MI). Nicotine outcomes show the highest baseline MI ($\approx$ 0.05–0.06), while illicit and prescription misuse groups fall roughly between 0.02 and 0.04. These MI values reflect stable associations between demographic or behavioral categories and broader substance-use patterns.

Under RAPPOR, however, MI collapses sharply. Across all groups and ε, RAPPOR MI drops to $\approx 10^{-3}$, and DP-segmentation often reduces it further into the 10^{-4} range. MI ratios (private MI relative to baseline MI) for the grouped outcomes remain extremely small, often below 0.05 and frequently close to 0.01. This confirms that local noise destroys most of the fine-grained statistical dependence between features and labels, even at the group level.

Nevertheless, resolution continues to modulate the tradeoff between MI and distortion. At lower resolutions ($K = 50$ or $K = 30$), MI remains small in absolute terms, but the substantial reduction in JS divergence implies that the remaining signal is more coherent and less overwhelmed by noise. This dynamic is reflected in higher privacy–utility scores at these intermediate-to-low resolutions.

Privacy-Utility Tradeoffs Across Groups: Examining F1, MI, and JS divergence jointly reveals that the overall privacy–utility behavior of grouped outcomes parallels the individual-drug case, albeit with more stable and more favorable numeric values. For the illicit and nicotine groups, increasing ε from 1 to 3 or 5 produces moderate increases in both F1 and MI while leaving JS divergence mostly unchanged; moving from $\varepsilon = 5$ to 10 produces only minor improvements. A similar pattern appears for the prescription-misuse group, where DP-segmentation at $K = 50$ or $K = 100$ consistently yields the strongest privacy–utility balance across all ε.

Importantly, even though baseline MI is higher in the grouped setting, the private models do not retain proportional amounts of that information. Instead, the improvement occurs because coarse-grained bins reduce distortion and produce more stable privatized features. This results in higher composite scores for moderate resolutions, particularly at $\varepsilon = 3$ and $\varepsilon = 5$.

Best-K and ε: Across the full set of group-level experiments, lower resolutions consistently delivered the strongest privacy–utility performance. For the illicit group, $K = 50$ repeatedly produced the highest composite scores across ε, while for nicotine and prescription misuse, the optimal range typically fell between $K = 30$ and $K = 100$. High-resolution settings ($K = 639$ and $K = 500$) rarely ranked near the top because their elevated distortion outweighed the modest MI advantages they offered.

With respect to ε, the global trend aligns with the individual-drug analyses: raising ε from 1 to 3 or 5 yields clear meaningful gains in F1 and MI for all groups, whereas further increasing ε to 10 provides diminishing returns. Crucially, the best-performing K values remain nearly unchanged across privacy budgets, indicating that segmentation resolution (not ε) is the primary determinant of privacy–utility efficiency

Effect of DP Segmentation: Across all privacy budgets and resolutions, label-aware DP segmentation consistently produced stronger privacy guarantees than RAPPOR alone while preserving essentially the same level of predictive utility. Aggregated across all drugs and all resolutions, DP segmentation reduced mutual information by a factor of approximately $7\times$ relative to RAPPOR at every privacy budget ($\varepsilon = 1, 3, 5, 10$). At $\varepsilon = 5$, for example, the mean mutual information dropped from 6.15×10^{-4} under RAPPOR to 8.7×10^{-5} under DP segmentation.

At the same time, global mean F1 values remained nearly unchanged. Across all drugs, the difference $\Delta\text{F1} = \text{F1}_{\text{DP-seg}} - \text{F1}_{\text{RAPPOR}}$ was slightly positive at $\varepsilon = 3$ and $\varepsilon = 5$, near zero at $\varepsilon = 10$, and effectively zero at $\varepsilon = 1$. This demonstrates that segmentation achieves substantial additional privacy protection without imposing a measurable penalty on predictive performance.

Taken together, these results show that DP segmentation improves the privacy–utility tradeoff not by increasing absolute accuracy, but by concentrating a small surviving signal into coarse, lower-distortion feature bins. This enables comparable predictive performance to RAPPOR while dramatically suppressing information leakage, establishing segmentation as a favorable structural enhancement to local differential privacy.

VI. Discussion

Across all experiments, a consistent theme emerged: under local differential privacy, predictive information is not carried by fine-grained categorical patterns but instead by coarse, population-level structure. RAPPOR's randomized-response noise destroys the high-resolution statistics that decision trees normally exploit: mutual information drops by one to two orders of magnitude, and the resulting privatized trees become unstable and heavily distorted at large resolutions. In contrast, when information is compressed into a small number of contiguous behavioral segments, the remaining signal becomes substantially more usable. This explains why F1 scores for RAPPOR-only models do not improve with increasing resolution K, while segmented models often achieve their best utility at moderate $K \in [30, 100]$.

The reason coarse structure succeeds is that segmentation aggregates many noisy categories into stable regions whose estimated frequencies and label-prevalence statistics are far less sensitive to RAPPOR fluctuations. At the category level, the decoded distribution is extremely spiky and unstable; small variations in bit counts can translate into dramatic changes in category probability. Once these categories are merged into

segments, however, variability is averaged out and the effective representation becomes both smoother and more robust. Because segmentation operates entirely on decoded aggregates, it does not weaken the underlying privacy guarantee: the full pipeline is a post-processing of the LDP mechanism. Thus, the partial recovery of predictive structure arises not from reversing privacy noise, but from reorganizing it into a form that remains statistically meaningful.

Another insight concerns the role of domain resolution relative to the privacy budget ε. Across all drugs and group outcomes, increasing ε beyond moderate levels provided only marginal benefits, whereas reducing resolution consistently decreased distortion and made the surviving signal easier to learn. This indicates that structural decisions—particularly how aggressively the domain is compressed—play a more decisive role in privacy–utility performance than tuning the privacy parameter alone. These findings reinforce a broader point from the differential privacy literature: when working under local privacy, representation design is a primary lever for enabling usable learning.

Finally, completing this project provided valuable intuition about the practical behavior of local differential privacy. While theoretical results often emphasize worst-case guarantees, the empirical work here showed how privacy mechanisms interact with real-world data distributions, class imbalance, and model structure. We observed firsthand that high-cardinality features are extremely fragile under LDP, that naive learning pipelines perform poorly on privatized inputs, and that structural recovery must occur above the level of individual categories. The segmentation framework we developed illustrates how meaningful prediction can still emerge from noisy data when the representation is thoughtfully designed. As such, these insights highlight both the limitations of LDP and the creative design space available for building systems that remain useful while preserving strong user-level privacy.

A. Strengths

This work offers several methodological and conceptual strengths. First, the study evaluates local differential privacy on a genuinely high-dimensional, real-world behavioral dataset, rather than on synthetic or overly simplified examples. This highlights the practical challenges of learning under LDP and demonstrates how severe the degradation can be when fine-grained categories are privatized at the client level.

Second, the proposed label-aware DP segmentation method is fully compatible with the post-processing property of differential privacy. Because segmentation operates only on RAPPOR-decoded aggregates, the method preserves the formal privacy guarantee while still enabling substantial improvements in structural stability and predictive utility. The approach is simple, transparent, and deployable within federated systems where servers never observe raw data.

Third, the evaluation goes beyond accuracy and incorporates distribution distortion, mutual information, and structural leakage metrics. These information-theoretic and topology-based analyses provide a more complete picture of privacy risk than performance metrics alone, aligning the study with emerging best practices in privacy-preserving machine learning.

Finally, the project contributes pedagogical value by demonstrating in a hands-on, empirical setting how differentially private mechanisms reshape data distributions, suppress information, and interact with standard learning algorithms. The insights gained through this process deepen understanding of LDP's practical constraints and illustrate how representation-level strategies can recover usable signal under strict privacy budgets.

B. Limitations & Ethics

Several limitations of this study must be acknowledged. First, all inference is performed on reconstructed population distributions rather than directly on individual client data. While this reflects realistic LDP deployment, it also means that segmentation accuracy is bounded by the quality of RAPPOR decoding. Errors in decoded distributions propagate downstream into segmentation and learning.

Second, our segmentation strategy assumes an ordered categorical domain after frequency sorting. While this works well for the engineered features used here, segmentation performance may degrade if category ordering lacks meaningful behavioral structure. More adaptive ordering strategies remain an area for future work.

Third, the NSDUH dataset is already legally anonymized and publicly released. As a result, our experiments do not capture the full ethical risks associated with deploying local privacy on raw, unprotected personal data. In real-world systems, demographic variables such as location, mental health, or socioeconomic status could pose much greater re-identification risks even under LDP.

From an ethical perspective, this work demonstrates both the power and the danger of privacy-preserving behavioral analytics. While RAPPOR and segmentation drastically reduce individual-level leakage, population-level predictions about sensitive behaviors such as drug use still carry societal risk. Any deployment of such systems must include safeguards against discrimination, misuse, and unfair targeting.

Finally, our work focuses on binary classification tasks. Extending these methods to continuous outcomes, causal inference, or time-series behavioral modeling presents significant additional challenges.

VII. Conclusion

This work examined whether meaningful predictive structure can be recovered from data protected by strong local differential privacy without compromising the underlying privacy guarantee. Using RAPPOR as a representative LDP mechanism and high-dimensional behavioral features derived from NSDUH, we demonstrated that local noise nearly eliminates direct feature–label dependence, as quantified by mutual information. However, we also showed that predictive utility did not vanish when learning operates on appropriately designed representations instead of raw privatized features.

The proposed label-aware DP segmentation method reshapes noisy population-level statistics into a small number of coarse, stable segments that remain fully protected by the post-processing immunity of differential privacy. Across a wide range of drugs, resolutions, and privacy budgets, segmentation preserved essentially the same predictive performance as RAPPOR alone while suppressing information leakage by an additional order of magnitude. These results establish that representation-level recovery—not denoising or privacy relaxation—is the key mechanism through which learning can occur under strict local privacy constraints.

Our experiments further revealed that feature resolution plays a more decisive role in governing privacy–utility trade-offs than the privacy budget itself. Coarse segmentations consistently reduced distributional distortion and produced the strongest overall performance, while increases in ε beyond moderate levels yielded only marginal improvements. This finding suggests that system designers should prioritize domain compression and structural stability alongside privacy parameter tuning when deploying locally private analytics.

More broadly, this study contributes to a growing body of evidence that privacy-preserving learning is fundamentally a problem of *representation*. When access to precise individual data is impossible by design, usable prediction must emerge from the careful organization of noisy aggregates into stable, low-complexity structures. The segmentation framework presented here provides a simple and principled example of how this can be achieved.

Future work may extend these ideas to sequential data, federated optimization, and heterogeneous feature spaces, as well as to fairness-aware segmentation strategies that explicitly constrain demographic disparities. As local differential privacy continues to gain relevance in industry and public policy, representation-level methods such as the one developed here will play an increasingly central role in enabling responsible, privacy-preserving machine learning.

Acknowledgment

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Appendix A
Post-Processing Privacy Guarantee of DP Segmentation

We formally justify that the proposed label-aware DP segmentation does not weaken the underlying local differential privacy guarantee provided by RAPPOR.

A. Statement of Guarantee

Let X denote the true client-side categorical value and let $\mathcal{M}_{\text{RAPPOR}}$ denote the randomized RAPPOR mechanism applied locally to each client. By definition, $\mathcal{M}_{\text{RAPPOR}}$ satisfies ε-local differential privacy:

$$\forall x, x', z: \quad \frac{\Pr[\mathcal{M}_{\text{RAPPOR}}(x) = z]}{\Pr[\mathcal{M}_{\text{RAPPOR}}(x') = z]} \leq e^{\varepsilon}.$$

Let $\mathcal{A}$ denote the server-side aggregation, decoding, and segmentation procedure mapping privatized reports into segment IDs. The full pipeline is therefore:

$$X \xrightarrow{\mathcal{M}_{\text{RAPPOR}}} Z \xrightarrow{\mathcal{A}} S.$$

B. Post-Processing Theorem

The post-processing immunity theorem of differential privacy states:

For any randomized mechanism $\mathcal{M}$ satisfying ε-DP and any (possibly randomized) function $\mathcal{A}$ independent of the raw private data, the composed mechanism $\mathcal{A} \circ \mathcal{M}$ also satisfies ε-DP.

C. Application to Our Pipeline

The segmentation algorithm operates *only* on decoded population-level statistics derived from RAPPOR outputs. It never accesses raw client categories and has no access to any unprotected user data. Therefore, segmentation is a valid post-processing function:

$$\mathcal{M}_{\text{seg}}(X) = \mathcal{A}(\mathcal{M}_{\text{RAPPOR}}(X)).$$

The post-processing immunity theorem of differential privacy states that for any randomized mechanism $\mathcal{M}$ satisfying ε-DP and any (possibly randomized) function $\mathcal{A}$ that does not depend on the raw private input directly, the composed mechanism $\mathcal{A} \circ \mathcal{M}$ also satisfies ε-DP.

D. Implication

This result is critical: segmentation can reshape structure but cannot leak additional information beyond what RAPPOR already permits. Therefore, any observed recovery in predictive utility arises from representation stability rather than privacy relaxation.

APPENDIX B
DESCRIPTIVE STATISTICS

TABLE III
DESCRIPTIVE STATISTICS FOR COARSENED FEATURE GROUPS

Age Group	Count	Percent (%)
12–17	34,284	19.73
18–34	70,198	40.41
35–49	37,504	21.59
50+	31,743	18.27
Sex		
Male	79,547	45.79
Female	94,182	54.21
Race Group		
White	99,992	57.56
Black	20,609	11.86
Other	20,216	11.64
Hispanic	32,912	18.94
Education Group		
HS or less	49,566	28.53
Some college	75,476	43.44
BA or more	48,687	28.02
Poverty Group		
Below poverty	32,234	18.55
Near poverty	34,583	19.91
Above poverty	106,912	61.54
K6 Distress Bin		
Low distress	38,816	22.34
Moderate distress	4,894	2.82
High distress	130,019	74.84

TABLE IV
PREVALENCE OF INDIVIDUAL DRUG OUTCOMES

Drug	**Users (Count)**	**Non-Users (Count)**	**Prevalence (%)**
marijuana	41,078	132,730	23.63
cocaine	3,363	170,445	1.93
crack	502	173,306	0.29
heroin	548	173,260	0.32
methamphetamine	1,493	172,315	0.86
inhalants	2,167	171,641	1.25
hallucinogens	6,528	167,280	3.76
tobacco	28,311	145,497	16.29
nicotine vaping	20,663	95,111	17.85
opioid misuse	38,645	135,163	22.23
tranquilizer misuse	17,140	156,668	9.86
sedative misuse	7,318	166,490	4.21
stimulant misuse	13,847	159,961	7.97
pain reliever misuse	38,529	135,279	22.17

TABLE V
PREVALENCE OF DRUG GROUPS

Drug Group	**Users (Count)**	**Non-Users (Count)**	**Prevalence (%)**
illicit	43,232	130,576	24.87
tobacco and nicotine	30,058	85,716	25.96
prescription misuse	55,525	118,283	31.95